AEP FROM ZERO • PART 1 • KNOW YOUR CUSTOMER (AEP)

EPISODE 05

AEP Data Ingestion Tutorial: Batch vs Streaming

A beginner's AEP data ingestion tutorial: batch vs streaming ingestion, and the one question that tells you which road a feed should take. Follow one row from source through Data Prep to the dataset and the Real-Time Customer Profile, map CSV columns to an XDM schema with the Map CSV to XDM schema workflow, and write Data Prep calculated fields that clean phone numbers, lower-case and trim emails, and derive a loyalty band. You will also meet source connectors, dataflows, Local file upload,

Previously: three empty blueprints

- Last time: three schemas, all Enabled for Profile
- Still empty: a blueprint is not a building
- Today: two messy CSV files and a live web feed
- Batch or streaming, Data Prep, three clean-ups
- Goal: reach the mapping screen with a plan

KEY IDEA

Bloom's three schemas are built and enabled for Profile, but empty. Today the real data arrives, messy, by two different roads.

Monday: two messy files and a web address

Feed	What arrived	The catch
CRM extract	10,000 customers	Casing, phones, duplicates
Order history	25,000 orders, 18 months	Quality never checked
Web events	Live clicks from the app	Sent only to an address
Trial files	500 rows each	Load small first

KEY IDEA

Ops hands over a dirty CRM extract and 18 months of orders, and the web team can only send live events to a web address.

Two roads in: batch and streaming

Question	Batch	Streaming
What moves	Files, on a schedule	Events, one at a time
Best for	History and bulk	Anything a journey reacts to
Lands in	Minutes or hours: fine	Seconds, by design
Bloom example	18 months of orders	Tonight's add-to-cart

KEY IDEA

Batch moves files on a schedule for history and bulk. Streaming sends events one at a time for anything a journey must react to.

Sorting Bloom's feeds onto two roads

Bloom feed	Road and reason
Nightly CRM export	Batch: bulk, on a schedule
Product catalogue, 200 items	Batch: changes rarely
Tonight's order, for a welcome email	Streaming: a journey reacts
Tonight's order, for a monthly report	Batch: nobody is waiting

KEY IDEA

The same kind of data can take either road. Who needs to react, and how fast, decides the road, not the data type.

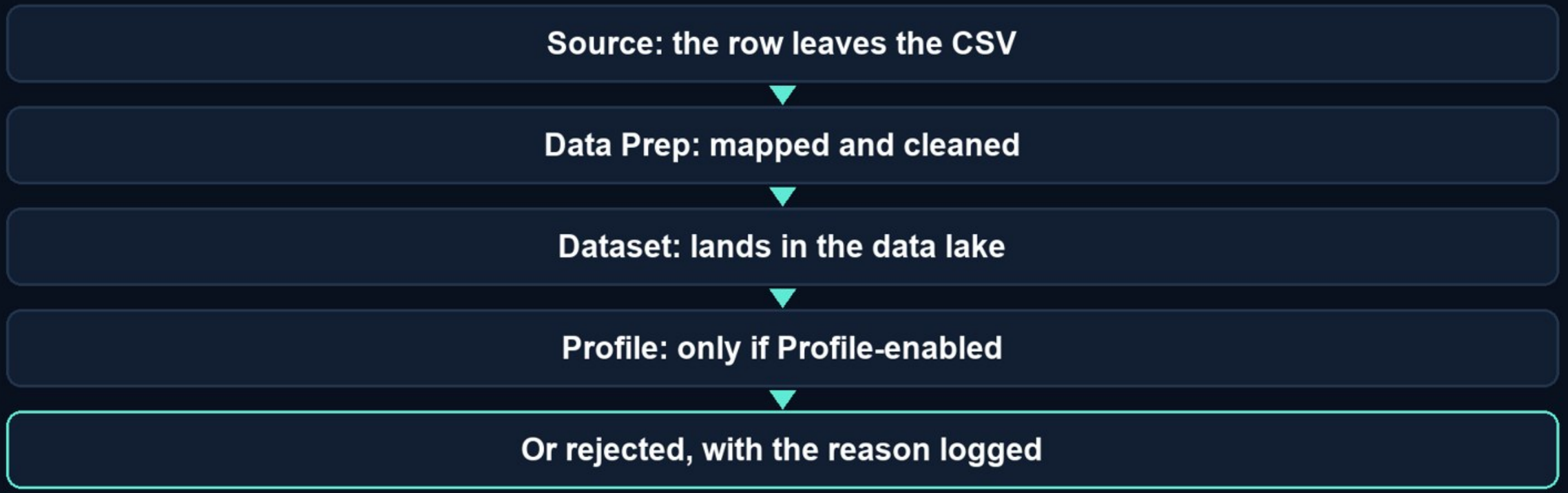
Sources, connectors and dataflows

- Source connector: a pre-built link into AEP
- Cloud storage, CRM, HTTP API, Local file upload
- Local file upload: CSV only, up to 1 GB
- Dataflow: source, through Data Prep, to a dataset
- Dataflow run: one execution, with its counts

KEY IDEA

A source connector is a pre-built link into AEP. A dataflow is the configured pipeline, and each dataflow run is what you monitor.

One row's path: profile or rejection



KEY IDEA

Every row passes source, Data Prep and dataset, and reaches the profile only if the dataset is enabled for Profile. The fifth ending is rejection.

What is Data Prep?

- Sits between the source and the dataset
- Job 1, mapping: each column to its XDM field
- email column to personalEmail.address
- Job 2, calculated fields: clean or derive values
- Never ingest dirt you can clean at the door

KEY IDEA

Data Prep sits between source and dataset. It maps each source column to an XDM field, and its calculated fields clean or derive values on the way in.

Planning the map for 14 columns

CSV column	Planned XDM home
email	personalEmail.address, the identity
first_name, last_name, phone	Person name, mobile phone number
loyalty_tier, points, join_date	_bloom loyalty fields
city, state, country	homeAddress fields
email, sms, push consent	_bloom consent fields

KEY IDEA

Every customer column gets a sensible home, and the three consent columns go to Bloom's own _bloom consent fields, because audiences will read them.

Why the identity columns matter most

File	bloom_id column	email column
Customer file	Plain attribute	Primary identity: Email
Order file	Primary identity: BloomID	Second identity: Email

KEY IDEA

Identity columns are how a row finds its person. A wrong attribute is a small error; a row with no identity is an orphan nobody can find.

Clean-up one: phones in five shapes

- (469) 555-0142 · 469.555.0142 · 4695550142
- 1-469-555-0142 · +1-469-555-0142
- Step 1: strip everything that is not a digit
- Step 2: 11 digits starting with 1? Drop that 1
- Step 3: add +1 once, so every row is +14695550142

KEY IDEA

A calculated field strips every non-digit and adds the +1 country code exactly once, so every phone lands as +14695550142.

Clean-up two: one person, one exact email

Arrives as	Lands as
' Priya.Sharma@Example.com '	priya.sharma@example.com
Capital letters anywhere	All lower case
Spaces at either end	Trimmed away

KEY IDEA

A calculated field lower-cases and trims every email, because the same person must always carry exactly the same identity value.

Clean-up three: a loyalty band from points

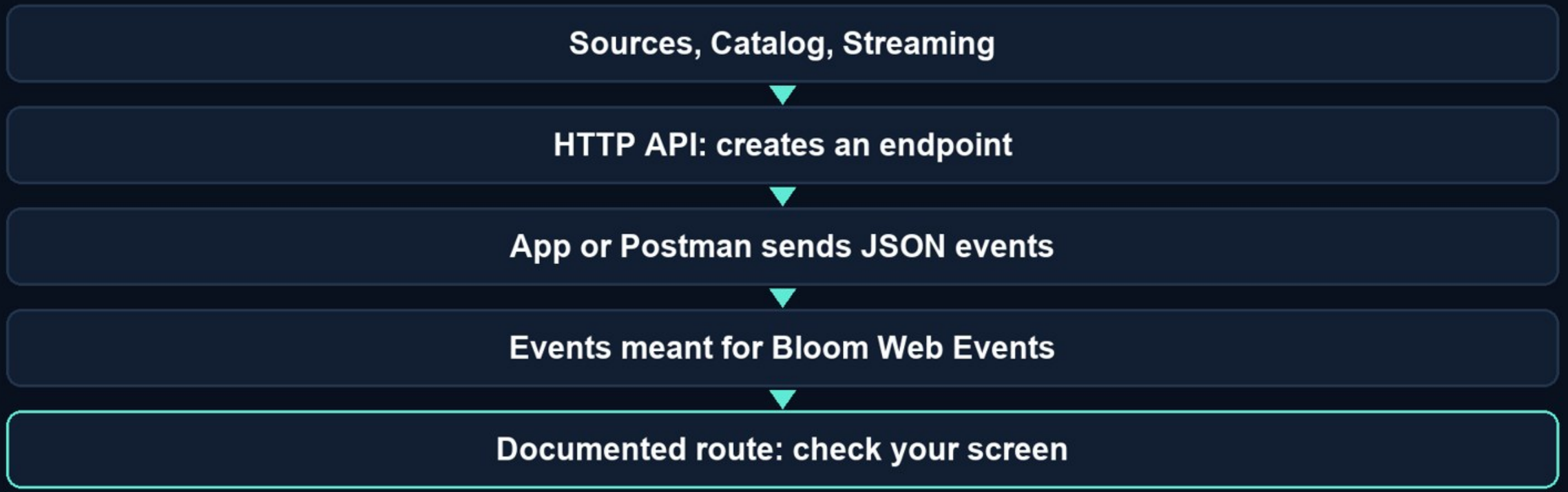
Points	Band	Why
2,340 (Priya)	Gold	1,500 or more
1,500 exactly	Gold	The boundary counts
1,499	Silver	500 to 1,499
35	Bronze	Below 500

```
loyalty_band = iif(points >= 1500, "Gold", iif(points >= 500, "Silver", "Bronze"))
```

KEY IDEA

The file has points but no band. A question inside a question derives Gold at 1,500 and above, Silver from 500, and Bronze below 500.

The streaming road: HTTP API and Postman



KEY IDEA

An HTTP API streaming source gives the web team an endpoint to send JSON events to, and Postman sends test events by hand. Check your own screen.

Into the product: three steps, three switches

Switch on Dataflow detail	What to do
Profile dataset	On, before you load anything
Error diagnostics	On, or failures stay unexplained
Enable partial ingestion	Off by default: see episode 6

KEY IDEA

Local file upload opens Map CSV to XDM schema: Dataflow detail, Select data, Mapping. The first step holds three switches that decide what happens later.

Select data, then the Mapping screen

- Select data: upload bloom-customers-500.csv
- Preview row one: Priya, Gold, 2,340 points
- Mapping: every column already auto-mapped
- Clean-ups: New field type, Add calculated field
- A full green bar: Mapped fields 14 of 14

KEY IDEA

The preview is the cheapest data-quality check you will ever get. Then the Mapping step opens with every column already guessed.

Interview corner

- “History and bulk go by batch.”
- “Anything a journey reacts to goes by streaming.”
- “Never ingest dirt you can clean at the door.”
- Proof: phone, email and band fixed in Data Prep
- Identity columns first: no identity, no person

KEY IDEA

Two lines that show engineering judgement: choose the road by reaction time, and clean dirt at the door, identity columns first.

Recap, your question, and the green bar

- Batch for history, streaming for reactions
- Source, Data Prep, dataset, profile, or rejected
- Clean at the door: phone, email, loyalty band
- Identity columns first; consent to _bloom
- Next: the failure lab

KEY IDEA

Choose the road by reaction time, clean at the door, map identities first. Next: what 14 of 14 mapped really means.